OAP200V

Object-oriented analysis, design and programming

00 – Introduction to the course

Teachers: Boban Vesin, Katerina Mangaroska

Boban Vesin

- Professor
- BSc, MSc, and PhD in Computer Science
  - Obtained at the University of Serbia, 2014
- Teaching courses for the past decade
  - Web application development, Data visualisation (USN)
  - Advanced software architecture (GU)
  - Web technologies (NTNU)
  - Informatikk prosjektarbeid II (NTNU)
  - On programming, web design, software engineering, etc. (Serbia)
- Programming experience in DMS Schneider electric Novi Sad

Katerina Mangaroska

- Associate professor
- Ph.D. in computer science
  - Obtained at the Department of Computer Science, NTNU, 2020
- Research interests
  - learning analytics
  - learning design
  - multimodal learning analytics
  - cognitive psychology
  - AI

Outline

- Overview of the course
- Course topics
- Course schedule
- Activities (assignment and exam)

Object-oriented programming – previous semester

- An introduction to object-oriented programming
- Understand concepts used in object-oriented programming.
- Know how to modify and create simple Java programs and have the tools to create more complex examples.
- Introduction to programming graphical user interfaces with an object-oriented language.
- Possess a fundamental knowledge of object-oriented programming, which will serve you well in advanced Java courses or in studying other programming languages such as Python, JavaScript, C++, C#, and Visual Basic.

This course

- The course introduces graphical modeling language and techniques for object-oriented analysis, design and documentation of information systems.
- The students will further develop their knowledge of object-oriented programming by creating larger applications.
- The course will provide skills in seeing the connection between models and program code, and understanding systems through model descriptions.
- The students will further develop their knowledge of object-oriented programming by creating larger applications.
- No formal prior knowledge is required, but the course is based on OBJ2000 Object-oriented Programming 1, or the equivalent.

Knowledge

- After completing the course, the student should have knowledge of:
  - how UML models can be used to show and understand the system from different perspectives
  - how UML models can be used for the analysis of user requirements and software design
  - modeling techniques in UML
  - the connection between UML syntax and specific programming code
  - software architecture
  - program design and architectural patterns
  - persistence: local file storage, serialization, and database access
  - streams, network classes, and I/O
  - threads and multitasking
  - user interface design
  - polymorphism and overloading
  - anonymous and inner classes
  - non-linear data structures

Skills

- After completing the course, the student will have the following skills:
  - create structure diagrams for simple and more complex systems
  - use UML to show software architecture and design
  - use UML to present dynamic and static elements of the software system
  - analyze existing software
  - model new software
  - visualise and plan software development process
  - use different classes for persistence
  - use textual and binary serialization
  - Program threads and perform multitasking
  - Program with non-linear data structures

General competence

- After completing the course, the student should be able to:
  - choose good perspectives for studying program systems
  - analyse relevant IT professional issues
  - have good expertise in object-oriented subjects
  - build larger object-oriented user programs
  - understand diagrams of larger structures and program design
  - reflect on your own professional practice and adjust this under the guidance

Timeline

Start of the course	17.8.2026
Delivery of the mandatory assignment	11.11.2026
Presentation the mandatory assignment	9.11.2026 and 11.11.2026
End of the semester	16.11.2026
Delivery of the final project and app	Exam date (to be determined)
Live project demo	Oral exam date (to be determined)
Feedback on the final exam:	Within three weeks

Java programming - Boban
Software design and analysis - Katerina
General information/Meetings - Boban and/or Katerina
Emil

Course schedule – Monday

Week	Day	Lecture topic (10:15-13:00)	Room	Labs (13:15-14:00)	Room
34	17.08.2026	Introduction to the course	C2-125		
35	24.08.2026	UI, Advanced Inheritance Concepts	C2-125	Labs	C2-125
36	31.08.2026	System Development, OO Paradigm, Object-Oriented Analysis (RA)	C2-125	Labs	C2-125
37	07.09.2026	Database access, Collections	C2-125	Labs	C2-125
38	14.09.2026	Collections, User Interface, DB, FileIO	C2-125	Labs	C2-125
39	21.09.2026	Project allocation and description, Running example	C2-125	Labs	C2-125
40	28.09.2026	Scrum meetings	C2-125	Labs	C2-125
41	05.10.2026	No lectures			
42	12.10.2026	Scrum meetings	K1-01	Labs	K1-01
43	19.10.2026	No lectures			
44	26.10.2026	Scrum meetings	K1-01	Labs	K1-01
45	02.11.2026	No lectures			
46	09.11.2026	Mandatory assignment presentation			K1-01
47	16.11.2026	Final exam guidelines	K1-01		

In a case of schedule update – most up-to-date version of the schedule will be in the Home page in Canvas

Java programming - Boban
Software design and analysis - Katerina
General information/Meetings - Boban and/or Katerina
Emil

Course schedule - Wednesday

Day	Lecture topic (10:15-13:00)	Room	Labs (13:15-14:00)	Room
19.08.2026	Eclipse IDE, Introduction to the running example, OO principles	D2-72		
26.08.2026	Exception Handling, File Input and Output	C3-97	Labs	C3-97
02.09.2026	Object-Oriented Design (CD)	F2-20	Labs	F2-20
09.09.2026	Object-Oriented Design (SD), Designing the UI and Deploying the New System	C3-99	Labs	C3-99
16.09.2026	Revision Control, Object-oriented design principles	C2-125	Labs	C2-125
23.09.2026	Running example	C1-65	Labs	C1-65
30.09.2026	Scrum meetings	K1-01	Labs	K1-01
07.10.2026	No lectures			
14.10.2026	Scrum meetings	K1-01	Labs	K1-01
21.10.2026	No lectures			
28.10.2026	Scrum meetings	K1-01	Labs	K1-01
04.11.2026	No lectures			
11.11.2026	Mandatory assignment presentation			K1-01
18.11.2026	No lectures			

In a case of schedule update – most up-to-date version of the schedule will be in the Home page in Canvas

Course topics

Introduction to the course
Mandatory assignment presentation
Final exam guidelines
Scrum meetings (3+1)

Software analysis and design (Katerina):

Introduction to System Development and Object-Oriented Paradigm
Fundamental Principles of Object-Oriented Analysis (RA)
Fundamental Principles of Object-Oriented Analysis (DM)
Fundamental Principles of Object-Oriented Design (CD)
Fundamental Principles of Object-Oriented Design (SD)
Designing the User-Interface and Deploying the New System

Programming (Boban):

Eclipse IDE, Introduction to the running example
Object-oriented design principles - intro, UI
Advanced Inheritance Concepts
Maven, Advanced arrays
Exception Handling
File Input and Output
Database access
Database access - II, Collections
User Interface
Object-oriented design principles - conclusion

Learning activities

- Lectures
- One mandatory assignment - presentation (in teams)
- Scrum meetings
- Final exam (in teams)

Course Workflow – Semester Structure – Part 1

- Theory & Practice (≈ 40% of semester)
  - Focus on *theoretical lectures* (analysis and design) and *practical programming lectures*.
  - All concepts will be linked to a **running example**: *Cinema Manager*.
  - The example will demonstrate the full process — problem analysis, system design, and implementation in Java.

Course Workflow – Semester Structure – Part 2

- Team Project (end of September onward)
  - Students will choose **their teams**.
  - Each team receives a **new problem description**.
  - Tasks: analyse the problem, design a solution, implement the application.
  - **Scrum meetings** with an assigned teacher to monitor progress and address challenges.
  - One **mandatory team presentation** to show mid-project progress to all teachers.

Scrum meetings

- Every two weeks, 30 minutes
- Each team will be assigned one teacher
- Purpose:
  - **Facilitate Communication:** Ensure everyone is on the same page regarding project progress.
  - **Identify Obstacles:** Highlight challenges early to address them promptly.
  - **Promote Accountability:** Each team member shares their progress and plans.
  - **Continuous Improvement:** Encourage reflection and adaptation for better team performance.
- Activities
  - Discuss what went well, what didn’t, and how to improve.
  - Define sprint goals

Course Workflow – Assessment and grading – Part 3

- One mandatory assignment (Must be approved before taking the exam)
  - Live presentation of the initial design - Fall 2026
  - Activities in ProTuS
  - No influence on the final grade
- The final exam
  - Delivered in November/December 2026 (on exam date)
- The oral final exam:
  - Usually, 10 days after final exam
- Evaluation based on design quality, implementation, code quality, and application of object-oriented concepts.

Work in teams

- All assignments and projects are delivered as a team
- Choose a team and
  - Send an email to me by Friday, 15.9.2026. 23:59
  - Team size:
    - minimum 5
    - maximum 6
  - The rest of the students will be randomly assigned to new groups
- Team members can have different final grade
  - The grade is determined by the individual contribution

Assessment and grading

- 15 credits
  - Grade is formed based only on the final exam
- “Pass” grade requirements:
  - Overall score > 50

Activity	Points
Final exam	100

Assessment and grading

- Grade scale A to F

Range	Grade
91-100	A
81-90	B
71-80	C
61-70	D
51-60	E
< 51	F

Course material

- All course material is (will be) uploaded in Canvas
- Material divided into modules (Home option)
  - schedule
- See Files for
  - Slides, material
- See Announcements for
  - Important news and updates, PM, Announcements
- See Assignments for
  - Description of assignments, submission guidlines, deadlines

Contact

- Office:
  - Room 136, Block C3 (it will be changed in october)
  - Campus Vestfold
- By email:
  - boban.vesin@usn.no
  - Katerina.Mangaroska@usn.no
- When contacting by email:
  - Use your official student email
  - Your full name must be in the FROM or the signature
  - Tag your subject line with [OAP200V]

Literature

Mandatory literature (Citations: 2, Pages: 1146)



BOOK **Java Programming**

Farrell, Joyce, Tenth edition, Boston, Massachusetts :, Cengage, 2022, Total Pages 654 sider

Add tags to item

Sent **Available** at Universitetet i Sørøst-Norge



BOOK **Introduction to systems analysis and design : an agile, iterative approach.**

Satzinger, John W.; Jackson, Robert B. (1961-), 6 utg. John W. Satzinger, Robert B. Jackson, Stephen D. Burd., Andover :, Cengage Learning, 2014., Total Pages 492 s. :

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Additional literature

Additional literature (Citations: 4, Pages: 823)



BOOK **Objektorientert programmering med Java : effektiv problemløsning i systemutvikling**

Holmstedt, Viggo, Bergen, Fagbokforl, cop. 2002, Total Pages 312 s.

Add tags to item

Sent Available at Universitetet i Sørøst-Norge

TEXT RESOURCE **Objektorientert analyse og design med UML**

Knut W. Hansson, 2013

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BOOK **Systems analysis and design in a changing world**

Satzinger, John W.; Jackson, Robert B.; Burd, Stephen D., 7th ed., Boston, Mass., Cengage Learning, 2016, Total Pages XXVII, 484 s.

Add tags to item

Sent Available at Universitetet i Sørøst-Norge



BOOK **Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development, Third Edition**

Craig Larman, Pearson, 2004-10-20

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Software

- Java Development Kit – JDK
- Development environment of your choice
  - Preferably Eclipse IDE
- Detailed guidelines for the installation are presented in Software chapter of the main course document

My advise for success

- Come to class
  - You have a better feeling of what is important
  - You spare time at home
- Do not postpone everything at the very end of the course
- Take advantage of the lab time, e.g., to ask questions
  - Feedback, feedback, feedback!
- Cheating will not be tolerated!

Questions?